4/3/23

§4.2 Variational characterizations of eigenvalues of hermitian matrices

<u>Theorem</u> (Rayleigh-Ritz)

Let $A \in M_n$ be hermitian with eigenvalues

$$\lambda_{min} := \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n =: \lambda_{max}$$

(recall $\sigma(A) \subseteq \mathbb{R}$)

Then

- $\lambda_1 x^* x \leq x^* A x \leq \lambda_n x^* x$, $\forall x \in \mathbb{C}^n$
- $$\lambda_{max} = \lambda_n = \max_{x \neq 0} \frac{x^* A x}{x^* x} = \max_{\|x\|_2 = 1} (x^* A x)$$
- $$\lambda_{min} = \lambda_1 = \min_{x \neq 0} \frac{x^* A x}{x^* x} = \min_{\|x\|_2 = 1} (x^* A x)$$

<u>Proof</u>

Proof

Since A is hermitian, $\exists$ unitary $U \in M_n$ & $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \in M_n(\mathbb{R})$, $\sigma(A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, $\lambda_1 \le \lambda_2 \le \ldots \le \lambda_n$, such that

$$A = U \Lambda U^*$$

Then $\forall x \in \mathbb{C}^n$,

$$x^* A x = x^* U \Lambda U^* x = (U^* x)^* \Lambda (U^* x)$$

$$= \sum_{j=1}^{n} \lambda_j (U^* x)_j^* (U^* x)_j$$

$$= \sum_{j=1}^{n} \lambda_j \overline{(U^* x)_j} (U^* x)_j$$

$$= \sum_{j=1}^{n} \lambda_j |(U^* x)_j|^2$$

Thus

$$\lambda_{min} \sum_{j=1}^{n} |(U^* x)_j|^2 \le x^* A x$$

$$= \sum_{j=1}^{n} \lambda_j |(U^* x)_j|^2 \le \lambda_{max} \sum_{j=1}^{n} |(U^* x)|^2$$

But $\displaystyle\sum_{j=1}^{n} |(U^* x)_j|^2 = \|U^* x\|_2^2 = \|x\|_2^2 = x^* x$

So $\boxed{\lambda_{min} x^* x \le x^* A x \le \lambda_{max} x^* x}$ #

The particular choice of x as

$$Ax = \lambda_{min} x,$$

shows

$$\lambda_{min} x^* x \le x^* A x = x^* \lambda_{min} x$$

$$\overset{\#}{\Longrightarrow} \quad \lambda_{min} = \min_{x \neq 0} \frac{x^* A x}{x^* x}$$

Similarly, $\lambda_{max} = \max_{x \neq 0} \frac{x^* A x}{x^* x}$

Replacing $x \longrightarrow \frac{x}{\|x\|_2}$ above, we get

$$\lambda_{min} = \min_{\|x\|_2 = 1} (x^* A x)$$

$$\lambda_{max} = \max_{\|x\|_2 = 1} (x^* A x)$$

$\blacksquare$

<u>Practical implication</u>

<u>Corollary</u> Let $A \in M_n$ be hermitian. Let $x \neq 0$. Let $\alpha = \frac{x^* A x}{x^* x}$

Then

$$\sigma(A) \cap (-\infty, \alpha] \neq \emptyset$$

$$\sigma(A) \cap [\alpha, \infty) \neq \emptyset$$

<u>Proof</u>

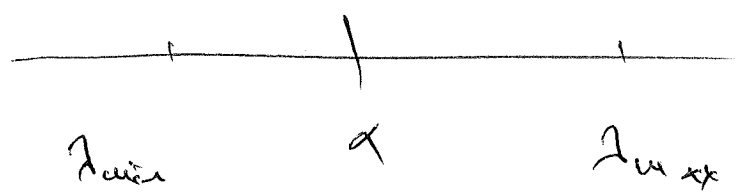
Clearly $\lambda_{min} \le \alpha \le \lambda_{max}$, where

$$\lambda_{min} = \min \sigma(A)$$
$$\lambda_{max} = \max \sigma(A) \quad \blacksquare$$

This provides an extremely easy & useful localization result for the spectrum of a hermitian matrix.

Given $x \neq 0$

Compute $\alpha = \dfrac{x^* A x}{x^* x}$



<u>Apply to general matrices:</u>

Bendixson's Theorem

Let $A \in M_n$, $Ax = \lambda x$, $x^* x = 1$

Let $H = \frac{A + A^*}{2}$, $S = \frac{A - A^*}{2}$ $(A = H + S)$

Then

$$\lambda = x^* A x = x^* H x + x^* S x$$

$$= \frac{x^* A x + x^* A^* x}{2} + \frac{x^* A x - x^* A^* x}{2}$$

$$= \frac{\lambda + \bar{\lambda}}{2} + \frac{\lambda - \bar{\lambda}}{2}$$

$$= \underbrace{\operatorname{Re} \lambda}_{\frac{\lambda + \bar{\lambda}}{2}} + i \underbrace{\operatorname{Im} \lambda}_{\frac{\lambda - \bar{\lambda}}{2}}$$

<u>But</u> if $\alpha_1 \le \alpha_2 \le \dots \le \alpha_n$ are the eigenvalues of H then by Rayleigh-Ritz applied to H:

$$\boxed{\alpha_1 \le \operatorname{Re} \lambda \le \alpha_n}$$

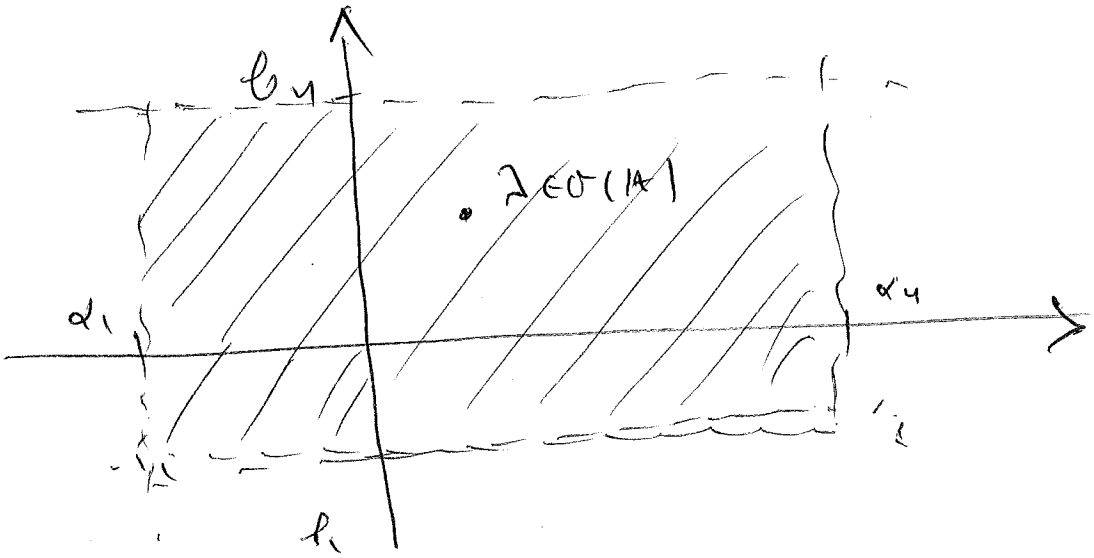
Similarly, if

$$\beta_1 \le \beta_2 \le \dots \le \beta_n$$

are the eigenvalues of $\frac{S}{i}$ (= hermitian)

$$\boxed{\beta_1 \le \operatorname{Im} \lambda \le \beta_n}$$

So we have Bendixson's bounding box:



$$A = H + S$$

$\sigma(H) = \{\alpha_1, \dots, \alpha_n\}$

$\sigma(S) = \{ib_1, \dots, ib_n\}$

$b_1 \le b_2 \le \dots \le b_n$

Back to hermitian matrices.

Apart / Analogously to Rayleigh-Ritz, can we (variationally) characterize all eigenvalues of a hermitian A (not just max & min)?

<u>Without proof</u>: If $\lambda_1 \le \lambda_2 \le \dots \le \lambda_n$ are the eigenvalues of hermitian A, then

$$\lambda_k = \min_{\substack{x \neq 0 \\ x \in \{u_1, \dots, u_k\}^\perp}} \frac{x^* A x}{x^* x},$$

where $u_1, \dots, u_n$ are the columns of a unitary U such that

$$A = U \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} U^*$$

$\{u_1, u_2, \dots, u_k\}^\perp$ = orthogonal complement of $\{u_1, \dots, u_k\}$

= all vectors in $\mathbb{C}^n$ that are orthogonal to u_j, $j = 1, 2, \dots, k$

Also

$$\lambda_{n-k} = \max_{\substack{x \neq 0 \\ x \in \{u_{n-k+1}, u_{n-k+2}, \dots, u_n\}^{\perp}}} \frac{x^* A x}{x^* x}$$

4/5/23

Let's actually look at the proof of extending Rayleigh-Ritz to λ_k:

Same set up: $A = A^*$, $A = U \Lambda U^*$ etc.

Consider $x \neq 0$, $x \perp u_1$ $\quad (U = [u_1 u_2 \cdots u_n])$

Then

$$x^* A x \overset{\text{as before}}{=} \sum_{j=1}^{n} \lambda_j |(U^* x)_j|^2 = \sum_{j=1}^{n} \lambda_j |u_j^* x|^2$$

$$\overset{x \perp u_1}{=} \sum_{j=2}^{n} \lambda_j |u_j^* x|^2$$

$$\geq \lambda_2 \sum_{j=2}^{n} |u_j^* x|^2 = \lambda_2 \sum_{j=1}^{n} |u_j^* x|^2$$

$$= \lambda_2 (U^* x)^* (U^* x) = \lambda_2 \|U^* x\|_2^2$$

$$= \lambda_2 x^* x$$

For $x = u_2 \perp u_1$, we get equality. So

$$\lambda_2 = \min_{\substack{x \neq 0 \\ x \perp u_1}} \frac{x^* A x}{x^* x}$$

$x \in \{u_1\}^\perp$

Similarly,

$$\lambda_k = \min_{\substack{x \neq 0 \\ x \in \{u_1, u_2, \ldots, u_{k-1}\}^\perp}} \frac{x^* A x}{x^* x}$$

■

<u>Natural question</u>: Are these Rayleigh-Ritz estimates for $\lambda_2, \ldots, \lambda_{n-1}$ any useful? This is because they seem to assume our knowledge of the columns of U. If we know them, we know the eigenvectors of A & so we know the eigenvalues. What's the point?

We can still use them $\longrightarrow$ Clever trick.

Let's take $w \in \mathbb{C}^n$ (fixed). Look at

$$\sup_{\substack{x^*x=1 \\ x \perp w}} (x^* A x) = \sup (x^* U \Lambda U^* x)$$

$$= \sup_{\substack{x^*x=1 \\ x \perp w}} \sum_{j=1}^{n} \lambda_j |U^* x|_j^2$$

$$\overset{\text{trick}}{=} \sup_{\substack{x^*x=1 \\ x = Uz \perp w}} \sum_{j=1}^{n} \lambda_j |z_j|^2 \qquad \left(\begin{array}{l} x = Uz \\ z = U^* x \end{array} \right)$$

$x \perp w$

$\Rightarrow z \perp w$

$$= \sup_{\substack{z^*z=1 \\ z \perp U^* w}} \sum_{j=1}^{n} \lambda_j |z_j|^2$$

$$\geq \sup_{\substack{z^*z=1 \\ z \perp U^* w \\ z_1 = z_2 = \cdots = z_{n-2} = 0}} \sum_{j=1}^{n} \lambda_j |z_j|^2$$

$$= \sup_{\substack{|z_{n-1}|^2 + |z_n|^2 = 1 \\ z \perp U^* w}} \left(\lambda_{n-1} |z_{n-1}|^2 + \lambda_n |z_n|^2 \right)$$

$$\geq \sup_{\substack{|z_{n-1}|^2+|z_n|^2=1 \\ z \perp U^* w}} \lambda_{n-1} \underbrace{\left(|z_{n-1}|^2+|z_n|^2\right)}_{=1} = \lambda_{n-1}$$

<u>So</u>

$$\sup_{\substack{x^* x=1 \\ x \perp w}} x^* A x \geq \lambda_{n-1}, \quad \forall w \in \mathbb{C}^n$$

Thus

$$\boxed{\lambda_{n-1} = \inf_{w \in \mathbb{C}^n} \sup_{\substack{x^* x=1 \\ x \perp w}} \left(x^* A x\right)}$$

This leads:

<u>Theorem</u> (Courant-Fischer)

Let $A \in M_n$ be hermitian with eigenvalues

$$\lambda_1 \le \lambda_2 \le \dots \le \lambda_n$$

and let $k \in \{2, \dots, n-1\}$. Then

$$\boxed{\lambda_k = \min_{w_1, w_2, \dots, w_{n-k} \in \mathbb{C}^n} \; \max_{\substack{x \neq 0,\, x \in \mathbb{C}^n \\ x \in \{w_1, w_2, \dots, w_{n-k}\}^\perp}} \left(\frac{x^* A x}{x^* x} \right)}$$

free of eigenvectors!

$$\boxed{\lambda_{n-k} = \max_{w_1, w_2, \dots, w_{k-1} \in \mathbb{C}^n} \; \min_{\substack{x \neq 0,\, x \in \mathbb{C}^n \\ x \in \{w_1, w_2, \dots, w_{k-1}\}^\perp}} \left(\frac{x^* A x}{x^* x} \right)}$$

typo in book "λ_k"

Practical application of Courant-Fischer:

<u>Theorem</u> (Weyl) Let $A, B \in M_n$ be hermitian, with eigenvalues

$$\lambda_1(A) \leq \cdots \leq \lambda_n(A)$$

$$\lambda_1(B) \leq \cdots \leq \lambda_n(B)$$

Consider the eigenvalues

$$\lambda_1(A+B) \leq \cdots \leq \lambda_n(A+B)$$

($A+B$ is indeed hermitian)

Then for $k = 1, 2, \ldots, n$,

$$\boxed{\lambda_k(A) + \lambda_1(B) \overset{\checkmark}{\leq} \lambda_k(A+B) \leq \lambda_k(A) + \lambda_n(B)}$$

Proof $\longrightarrow$

<u>Proof</u> $\forall x \in \mathbb{C}^n$

(Rayleigh-Ritz) $\lambda_1(B) \le \dfrac{x^* B x}{x^* x} \le \lambda_n(B)$

$$\Longrightarrow \lambda_k(A+B) \overset{C\text{-}F}{=} \min_{w_1,\dots,w_{n-k}} \max_{\substack{x \neq 0 \\ x \perp \{w_1,\dots,w_{n-k}\}^\perp}} \frac{x^*(A+B)x}{x^* x}$$

$$= \min \max \left[\frac{x^* A x}{x^* x} + \frac{x^* B x}{x^* x} \right]$$

$$\ge \min \max \left[\frac{x^* A x}{x^* x} + \lambda_1(B) \right]$$

$$= \underbrace{\min \max \left[\frac{x^* A x}{x^* x} \right]} + \lambda_1(B)$$

$$\overset{C\text{-}F}{=} \lambda_k(A) + \lambda_1(B)$$

Similarly the other inequality ∎

4/7/23

Sylvester Law of Inertia

Set-up: Let $A \in M_n$ be hermitian.

The <u>inertia</u> of A is

$$i(A) = \left(i_+(A), i_-(A), i_0(A) \right),$$

where $i_+(A) = \#$ of positive eigenvalues of A

$i_-(A) = \#$ of negative ——— " ———

$i_0(A) = \#$ of zero ——— " ———

We know $\exists$ unitary U, diagonal Λ such that

$$A = U \Lambda U^*, \quad \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$$

WLOG, if $i(A) = (\alpha, \beta, \gamma)$, we order the eigenvalues so that

$$\lambda_1, \lambda_2, \ldots, \lambda_\alpha > 0$$

$$\lambda_{\alpha+1}, \lambda_{\alpha+2}, \ldots, \lambda_{\alpha+\beta} < 0$$

$$\lambda_{\alpha+\beta+1} = \lambda_{\alpha+\beta+2} = \cdots = \lambda_n = 0$$

Set

$$D := \operatorname{diag}\left(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_\alpha}, \sqrt{-\lambda_{\alpha+1}}, \ldots, \sqrt{-\lambda_{\alpha+\beta}}, 1, \ldots, 1 \right)$$

$$\in M_n$$

Notice

$$\Lambda = D \underbrace{\begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & -1 & & & & & \\ & & & & \ddots & & & & \\ & & & & & -1 & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{bmatrix}}_{I(A)} D \quad ;$$

we call $I(A)$ the <u>inertia matrix</u> of A

So $A = U \Lambda U^* = S\, I(A)\, S^*$, where $S = UD$

In other words, "every hermitian matrix is $*$-congruent to a real diagonal matrix"

We will see, very much like A & $I(A)$ have the same inertia, <u>all</u> $*$-congruent hermitian matrices share the same inertia.

<u>Theorem</u> (Sylvester Law of Inertia)

Let $A, B \in M_n$ be hermitian. Then there exists invertible $S \in M_n$ such that

$$A = SBS^*$$

if and only if $i(A) = i(B)$.

Proof

($\Leftarrow$) Suppose $i(A) = i(B)$. Then

$$A = S_A \, I(A) \, S_A^* \quad (\text{some } S_A \text{ invertible})$$

$$B = S_B \, I(B) \, S_B^* \quad (\text{— } S_B \text{ —})$$

However, $I(A) = I(B)$. So

$$B = S_B \, I(A) \, S_B^* = S_B \left(S_A^{-1} A (S_A^{-1})^* \right) S_B^*$$

$$= (S_B S_A^{-1}) A ((S_A^{-1})^* S_B^*)$$

$$= \underbrace{(S_B S_A^{-1})}_{\text{invertible}} A (S_B S_A^{-1})^*$$

i.e. B & A are $*$-congruent.

($\Rightarrow$) Conversely, suppose $A = SBS^*$, S invertible.

Since $\operatorname{rank}(A) = \operatorname{rank}(SBS^*) = \operatorname{rank}(B)$

$$\Rightarrow \quad i_+(A) + i_-(A) = i_+(B) + i_-(B)$$

$$\Rightarrow \quad i_0(A) = i_0(B)$$

To show $i(A) = i(B)$, it suffices to show

$$i_+(A) = i_+(B)$$

Let $u_1, u_2, \dots, u_\alpha$ be the orthonormal eigenvectors corresponding to $\lambda_1, \lambda_2, \dots, \lambda_\alpha \in \sigma(A)$, $\alpha = i_+(A)$.

If $x = \sum_{i=1}^{\alpha} t_i u_i$, $t_i \in \mathbb{C}$, then

$$x^* A x = \lambda_1 |t_1|^2 + \dots + \lambda_\alpha |t_\alpha|^2 \quad (x \neq 0) > 0$$

(compute $x^* A x$ & use $u_i^* u_j = 0$ if $i \neq j$, $u_i^* u_i = 1$)

Thus

$$x^* A x = \underbrace{x^* S B S^* x} = (S^* x)^* B (S^* x) > 0$$

for all nonzero $x \in \operatorname{span}\{u_1, \dots, u_\alpha\}$.

Or, equivalently,

$$y^* B y > 0 \quad \text{for all nonzero } y \in \operatorname{span}_{1 \le i \le \alpha}\{S^* u_i\}$$

i.e. $y^* B y > 0$, $\forall$ nonzero vectors y in

$$W = \operatorname{span}_{1 \le i \le \alpha}\{S^* u_i\}.$$

But

$$\dim W = \alpha$$

<u>Claim</u> (we'll prove after) This implies, $i_+(B) \ge \alpha = i_+(A)$

Reversing the roles of A & B (equivalence relation), we have

$$i_+(A) \geq i_+(B)$$

and so $i_+(A) = i_+(B)$ ∎

To justify the "claim" in the proof above let's prove the following impressive result:

<u>Theorem</u> Let $A \in M_n$ be hermitian with eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$.

Let S_k be a subspace of $\mathbb{C}^n$ such that

$$0 < \dim S_k = k \leq n.$$

If $\exists\, c_2 \in \mathbb{R}$ such that

$$\boxed{x^* A x \geq c_2\, x^* x}, \quad \forall x \in S_k,$$

then

$$\boxed{\lambda_n \geq \lambda_{n-1} \geq \dots \geq \lambda_{n-k+1} \geq c_2}$$

If $\exists\, c_1 \in \mathbb{R}$ such that

$$\boxed{x^* A x \leq c_1\, x^* x}, \quad \forall x \in S_k,$$

then

$$\boxed{c_1 \geq \lambda_k \geq \lambda_{k-1} \geq \dots \geq \lambda_1}$$